🤖 *RationGuard – AI Module Overview*

🌐 *Overall Purpose*

- ***Detect fraud and anomalies in government ration distribution in real-time.***
- ***Ensure eligible beneficiaries receive correct ration quantities.***
- ***Assist officials by highlighting suspicious transactions and centers.***
- ***Provide actionable insights for reporting and corrective measures.***

Core AI Functionalities

- ***Duplicate Beneficiary Detection: Identifies multiple entries for the same person or ration card.***
- ***Ghost Beneficiary Detection: Flags transactions for non-existent beneficiaries or fake entries.***
- ***Quantity Anomaly Detection: Finds mismatches in distributed vs. allocated ration quantities.***
- ***Location/Center Analysis: Detects irregular patterns or unusual distribution at specific centers.***
- ***Trend Analysis: Monitors daily/weekly/monthly trends to detect spikes in fraud activity.***
- ***Confidence Scoring: Each anomaly gets a probability score (e.g., 95% chance of duplicate card).***
- ***Explainability: AI provides reasoning for each flagged transaction (e.g., "Duplicate card ID", "Exceeds average allocation").***

⚙️ *Data Input & Preprocessing*

- ***Uses datasets of beneficiaries, centers, transactions, and historical anomalies.***
- ***Cleans and normalizes data (removes duplicates, handles missing values).***
- ***Converts categorical variables (district, ration type) into model-readable format.***
- ***Optional synthetic data generation for testing/demo purposes.***

🧮 *Models & Techniques*

- ***Rule-Based Detection: Quick initial checks for duplicate IDs, wrong quantity ranges,***

and blacklisted centers.

- ***Machine Learning Models (Optional Advanced Demo):***
  - ***Decision Trees / Random Forests → Detect unusual patterns in historical transactions.***
  - ***Clustering (K-Means) → Identify outlier centers or beneficiaries.***
  - ***Anomaly Detection Algorithms (Isolation Forest, Local Outlier Factor) → Find rare fraudulent events.***
- ***Integration: Python AI module can be called via REST API from the Node.js backend.***

Output & Integration

- ***Flags suspicious transactions and stores them in Alerts collection in MongoDB.***
- ***Provides data to frontend dashboard:***
  - ***Overview cards (Total fraud, active alerts)***
  - ***Live transaction table highlighting anomalies***
  - ***Heatmap showing fraud-prone centers***
  - ***Confidence score and reasoning for each anomaly***

Analytics & Reporting

- ***Provides aggregated analytics for officials:***
  - ***Fraud distribution by district/center***
  - ***Types of anomalies detected***
  - ***Daily/weekly trends of suspicious activity***
- ***Supports exportable reports (PDF/CSV) for audits and verification.***

Optional Advanced Features

- ***AI continuously retrains on new data to improve detection accuracy.***
- ***Detects evolving patterns or new types of fraud (adaptive AI).***
- ***Simulated "What-if" scenarios: Officials can test potential changes in ration distribution and see AI's predicted outcomes.***